Khidmat Report: Foundations of Python

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The objective of this project was to provide students with a solid foundation in core Python programming concepts, subsequently introducing key libraries for data manipulation and visualization for Baitussalam Welfare Trust.

Baitussalam Welfare Trust was the client for this project.

The "Foundations of Python" was a five-week program designed to build strong foundational programming skills in Python before introducing essential libraries for data handling and visualization. Initially conceived as a course focused on scientific computing, an assessment of the students' existing knowledge led us to dedicate significant time to reinforcing core Python concepts. The program ultimately covered foundational programming principles, essential data structures, error handling, and introduced key libraries such as Pandas and NumPy, alongside basic plotting techniques, providing a stepping stone towards data science applications.

The program was conducted in person, with three weekly sessions: two 3-hour lectures (with in-class practice questions) on Mondays and Wednesdays, and a 3-hour lab on Saturdays. Lecture sessions introduced topics and provided small exercises to reinforce learning. Lab sessions focused on extensive hands-on practice, where participants applied weekly topics through coding exercises and practical data manipulation tasks. Both of us were involved at all times to managed the sessions and maximize student learning outcomes.

Week 0: January 6, 2025 - January 12, 2025

This preliminary week was dedicated to understanding student needs. We conducted 100 interviews, totaling 5 hours, to assess current Python proficiency and tailor the course content effectively.

Day	Activity	Topics
Throughout the week	Student Interviews	Conducted 100 student interviews to assess current understanding and inform course design.

Activity	**Time in Hours**	**Individual Breakdown**
Throughout the week		
Conducting student interviews and analysis	4.0	Hussain
Conducting student interviews and analysis	4.0	Burhanuddin

Summary - Week 0	**Hussain (Hours)**	**Burhanuddin (Hours)**	**Total (Hours)**
Weekly Total	4.0	4.0	8.0

Week 1: January 13, 2025 - January 19, 2025

This week focused on establishing a strong foundation in core Python programming concepts, including variables, data types, control flow, and basic data structures, reinforced through hands-on lab exercises and subsequent review of submitted work.

Day	Activity	Topics
Lecture 1	Lecture and Exercises	Python setup and introduction. Core Python basics: Data types, variables, and basic input/output.
Lecture 2	Lecture and Exercises	Control flow: Conditionals (if/else), Loops (for, while). Introduction to Functions. Data Structures: Lists and Tuples.
Saturday	Lab Assignment	Reinforcing Python basics: Practicing with variables, data types, conditionals, loops, functions, lists, and tuples through coding exercises. Basic file handling.

Activity	**Time in Hours**	**Individual Breakdown**
Monday		
Lecture content preparation	2.0	Hussain
In class assignment preparation	2.0	Burhanuddin
Lecture	1.5	Hussain
In class practice questions (explaining and leading)	1.5	Burhanuddin
In class practice questions (assisting students with questions and issues)	1.5	Hussain
Wednesday		
Lecture content preparation	2.0	Burhanuddin
In class assignment preparation	2.0	Hussain
Lecture	1.5	Burhanuddin
In class practice questions (explaining and leading)	1.5	Hussain
In class practice questions (assisting students with questions and issues)	1.5	Burhanuddin
Saturday		
Building the lab	4.0	Hussain
Conducting the lab	3.0	Burhanuddin
Helping students in lab	3.0	Hussain
Checking submitted labs	2.0	Hussain
Checking submitted labs	2.0	Burhanuddin

Summary - Week 1	**Hussain (Hours)**	**Burhanuddin (Hours)**	**Total (Hours)**
Weekly Total	17.5	13.5	31.0

Week 2: January 20, 2025 - January 26, 2025

This week reinforced core Python concepts from Week 1, introduced error handling with exceptions, and covered fundamental data structures like lists and dictionaries, with all new concepts reinforced through practical exercises in the lab and feedback on submissions.

Day	Activity	Topics
Lecture 1	Lecture and Exercises	Reinforcing Week 1 concepts: Variables, conditionals, functions, and loops. Introduction to Error Handling and Exceptions.
Lecture 2	Lecture and Exercises	Data Structures: Introduction to lists, dictionaries, sets and tuples.
Saturday	Lab Assignment	Applying learned concepts. Internal workings of lists, dictionaries, sets, and tuples. Mutability and immutable objects in Python. Understanding that everything in Python is an object.

Activity	**Time in Hours**	**Individual Breakdown**
Monday		
Lecture content preparation	2.0	Burhanuddin
In class assignment preparation	2.0	Hussain
Lecture	1.5	Burhanuddin
In class practice questions (explaining and leading)	1.5	Hussain
In class practice questions (assisting students with questions and issues)	1.5	Burhanuddin
Wednesday		
Lecture content preparation	2.0	Hussain
In class assignment preparation	2.0	Burhanuddin
Lecture	1.5	Hussain
In class practice questions (explaining and leading)	1.5	Burhanuddin
In class practice questions (assisting students with questions and issues)	1.5	Hussain
Saturday		
Building the lab	4.0	Burhanuddin
Conducting the lab	3.0	Hussain
Helping students in lab	3.0	Burhanuddin
Checking submitted labs	2.0	Hussain
Checking submitted labs	2.0	Burhanuddin

Summary - Week 2	**Hussain (Hours)**	**Burhanuddin (Hours)**	**Total (Hours)**
Weekly Total	13.5	17.5	31.0

Week 3: January 27, 2025 - February 3, 2025

The final week focused on introducing NumPy and revising Pandas with advanced operations, conducting a comprehensive review of all learned concepts through practice questions, and providing guidance on future steps in scientific computing and data science careers. An optional project was also introduced for further practice.

Day	Activity	Topics
Lecture 1	Lecture and Exercises	Introduction to NumPy: Foundational operations, emphasizing speed and comparison with Python lists. Introduction to Pandas basics, understanding Pandas objects in terms of dictionaries and lists, and exploring similar native methods.
Lecture 2	Lecture and Exercises	Comprehensive revision of all learned concepts with practice questions. Discussion on next steps: further learning resources, importance of LinkedIn, and desired competencies in the field.
Saturday	No Class	Optional Project Introduction: A project with multiple bite-sized pieces and commented-out code for students to complete. An explainer guide was provided on YouTube.

Activity	**Time in Hours**	**Individual Breakdown**
Monday		
Lecture content preparation	2.0	Hussain
In class assignment preparation	2.0	Burhanuddin
Lecture	1.5	Hussain
In class practice questions (explaining and leading)	1.5	Burhanuddin
In class practice questions (assisting students with questions and issues)	1.5	Hussain
Wednesday		
Lecture content preparation	2.0	Burhanuddin
In class assignment preparation	2.0	Hussain
In class practice questions (explaining and leading)	1.5	Hussain
In class practice questions (assisting students with questions and issues)	1.5	Burhanuddin
Optional Project		
Building commented code	4.0	Burhanuddin
Creating explainer video	2.0	Burhanuddin

Summary - Week 3	**Hussain (Hours)**	**Burhanuddin (Hours)**	**Total (Hours)**
Weekly Total	8.5	11.5	20.0

Conclusion

The project, "Foundations of Scientific Computing and Data Science in Python," for Baitussalam Welfare Trust involved an initial assessment week (Week 0) followed by a three-week instructional program. During Week 0, we conducted 100 student interviews to gauge their existing Python knowledge, which significantly informed the course design. Over the subsequent three instructional weeks (Weeks 1-3), we covered foundational programming concepts, key scientific libraries like NumPy and Pandas, and essential hands-on computational techniques through lectures, exercises, lab assignments, and feedback on submitted work. The program was conducted in-person with a focus on interactive learning and practical application.

Overall Project Summary	Hussain (Hours)	Burhanuddin (Hours)	Total (Hours)
Project Total	43.5	46.5	90.0

Resources

- CS50's Introduction to Programming with Python by Harvard University: `https://cs50.harvard.edu/python/2022/weeks/0/`
- CS50 Week 3 Project Final Guidelines on YouTube: `https://www.youtube.com/watch?v=RlJKxm7Q_DM`
- *Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython* by Wes McKinney: `https://www.amazon.com/Python-Data-Analysis-Wrangling-IPython/dp/1491957662`
- GitHub Repository for Project Code and Material – Khidmat by Burhanuddin: `https://github.com/burhanuddin6/Khidmat`
- Project for students interested in going beyond: `https://github.com/burhanuddin6/scwp_project`

Khidmat Completion Form

To be completed by the external supervisor.

Please use the space below to provide any comments you may have on the students' performance, the Khidmat program, or any other feedback you want to share with Habib University's Khidmat committee. We can also be reached at khidmat@sse.habib.edu.pk.

I hereby certify that I supervised Hussain Mustansir and Burhanuddin Ezzi for the Khidmat described in this report. Furthermore, that I have read and agree with the weekly updates included in this report. My signature below marks the successful completion of the work undertaken for the Khidmat.

Danish

Name and signature

Clifton block 4 opposite dolmen mall

Location and date